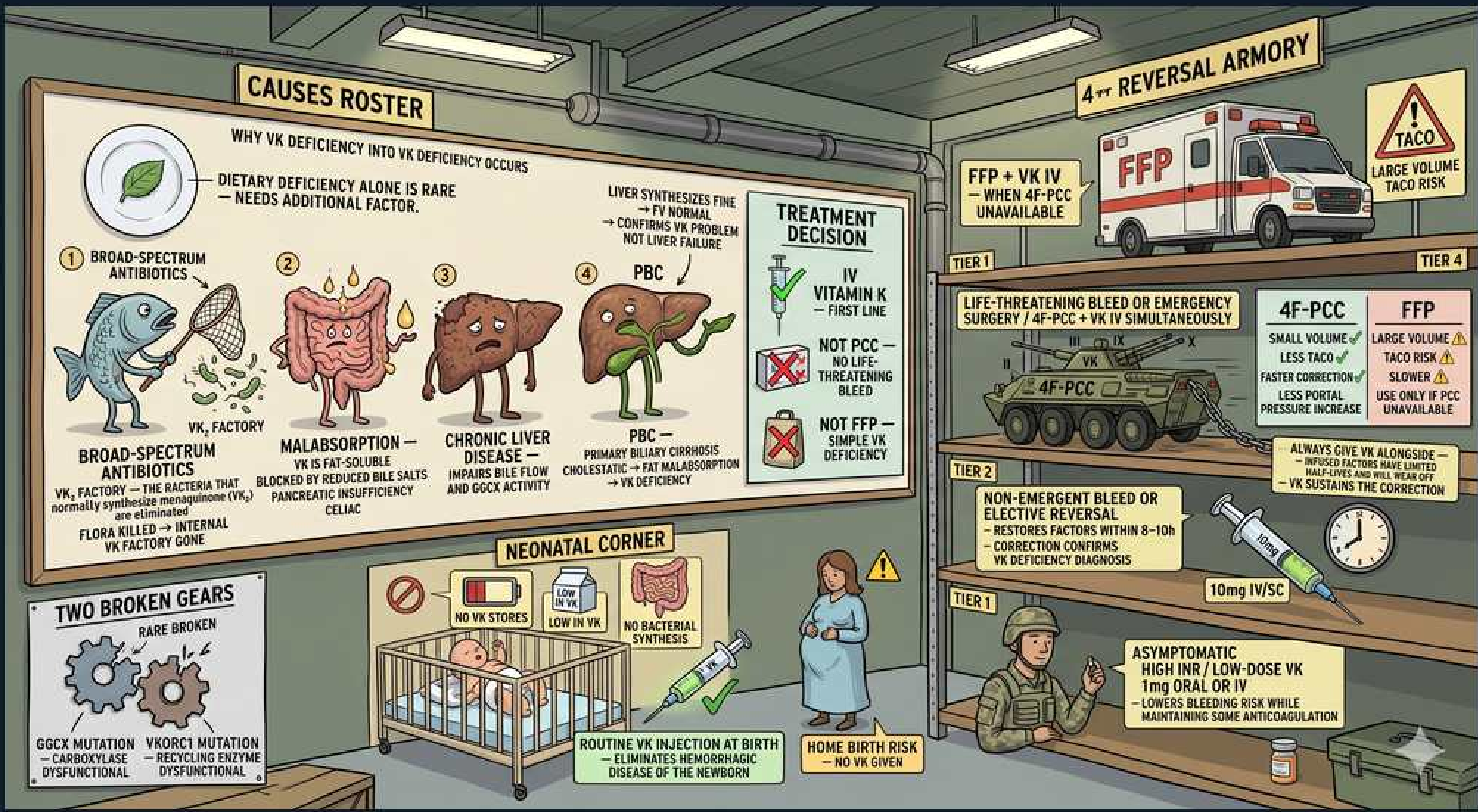
Coagulation — Vitamin K Deficiency (causes, newborn, reversal)



1. Dietary deficiency alone is rare

WHAT YOU SEE

Causes roster leaf icon 'dietary deficiency alone is rare, needs additional factor'

WHAT IT ENCODES

Pure dietary vitamin K deficiency is UNCOMMON in adults because (1) leafy greens supply vitamin K1 (phylloquinone) and (2) gut bacteria synthesize vitamin K2 (menaquinone). Clinically significant deficiency usually requires a SECOND hit — malabsorption, broad-spectrum antibiotics, prolonged NPO/poor intake, or warfarin. Mnemonic: 'diet alone rarely does it — you need a second hit.'

BOARD TRAP

A healthy diet does not protect against deficiency if there is malabsorption or antibiotic-mediated flora loss — and a 'normal diet' history does not rule deficiency out.

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2. Broad-spectrum antibiotics

WHAT YOU SEE

Cause #1 net-catching gut flora 'broad-spectrum antibiotics kill flora'

WHAT IT ENCODES

Broad-spectrum antibiotics kill the colonic flora that produce vitamin K2, precipitating deficiency — especially in a hospitalized, NPO, or poorly-eating patient (loss of both intake and synthesis). Cephalosporins with an N-methylthiotetrazole (NMTT) side chain (e.g., cefoperazone, cefotetan) ALSO directly inhibit the vitamin K cycle, a double hit. Mnemonic: 'antibiotics net the gut bugs that make K.'

BOARD TRAP

This is loss of an INTERNAL (flora) source, not reduced dietary intake. The combination of antibiotics + NPO + critical illness is the classic inpatient setup for a rising INR.

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3. Malabsorption (fat-soluble)

WHAT YOU SEE

Cause #2 'malabsorption, fat-soluble, reduced bile salts, pancreatic insufficiency, celiac'

WHAT IT ENCODES

Vitamin K is FAT-SOLUBLE, so any fat-malabsorption state lowers it: bile salt deficiency/cholestasis, pancreatic exocrine insufficiency, celiac disease, IBD, short-bowel syndrome. Often accompanied by deficiency of the other fat-soluble vitamins A, D, E. Mnemonic: 'no fat absorption = no ADEK.'

BOARD TRAP

Cholestasis blocks vitamin K ABSORPTION even when intake is adequate — and ORAL vitamin K may fail in cholestasis (no bile to absorb it), so give it parenterally.

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4. Chronic liver disease

WHAT YOU SEE

Cause #3 'chronic liver disease' liver icon

WHAT IT ENCODES

Chronic liver disease causes a DOUBLE problem: reduced bile → vitamin K malabsorption (a correctable deficiency component) AND impaired hepatocyte synthesis of factors (not correctable by vitamin K). A vitamin K trial helps tease the two apart. Mnemonic: 'liver = bad absorption AND bad factory.'

BOARD TRAP

If the PT FAILS to correct after vitamin K, suspect hepatic SYNTHETIC failure (factor V will be low), not simple deficiency. Full correction = deficiency; partial/no correction = synthetic failure.

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5. PBC / cholestasis

WHAT YOU SEE

Cause #4 'PBC, primary biliary cirrhosis, cholestatic, fat malabsorption'

WHAT IT ENCODES

Cholestatic diseases — primary biliary cholangitis (PBC), primary sclerosing cholangitis, biliary obstruction — impair bile flow → fat-soluble vitamin (A, D, E, K) malabsorption → coagulopathy. Because the defect is absorptive, it responds to PARENTERAL vitamin K. Mnemonic: 'no bile → no fat-soluble vitamins.'

BOARD TRAP

In cholestasis ORAL vitamin K may fail (no bile salts to solubilize it); give IV/SC (parenteral) vitamin K instead.

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6. Two broken gears (GGCX/VKORC1)

WHAT YOU SEE

Bottom-left 'two broken gears, rare GGCX / VKORC1 mutation, warfarin resistance'

WHAT IT ENCODES

Rare HEREDITARY defects of the vitamin K machinery: loss-of-function mutations in GGCX or VKORC1 cause combined vitamin K–dependent factor deficiency, while certain VKORC1 variants cause WARFARIN RESISTANCE (requiring unusually high doses). Mnemonic: 'broken gears in the carboxylation machine.'

BOARD TRAP

These are rare GENETIC mimics, not acquired deficiency — suspect when warfarin dosing is bizarrely high (resistance) or in a congenital combined-factor-deficiency vignette.

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7. Newborn — no gut flora

WHAT YOU SEE

Neonatal corner crib 'no stores, low in milk, no bacterial synthesis'

WHAT IT ENCODES

Newborns are physiologically vitamin K deficient: minimal hepatic stores, poor transplacental transfer, a STERILE gut (no flora to make K2), and breast milk is LOW in vitamin K. This causes vitamin K deficiency bleeding (VKDB), formerly hemorrhagic disease of the newborn — including catastrophic intracranial hemorrhage. Mnemonic: 'sterile gut + low-K milk = bleeding baby.'

BOARD TRAP

BREASTFEEDING (low vitamin K) INCREASES VKDB risk relative to fortified formula — a classic exam point. Late VKDB (2–12 weeks) presents with intracranial bleeding.

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8. Vitamin K IM at birth

WHAT YOU SEE

'Routine VK injection at birth eliminates hemorrhagic disease of the newborn'

WHAT IT ENCODES

A single routine INTRAMUSCULAR vitamin K injection at birth prevents both early and late VKDB and is standard of care. Oral regimens are less reliable (especially for late VKDB). Mnemonic: 'one IM shot at birth = no newborn bleed.'

BOARD TRAP

Parental REFUSAL of the IM injection (often with home birth) leaves the infant at risk for life-threatening VKDB/intracranial hemorrhage — a recognized board scenario.

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9. Vitamin K is first-line

WHAT YOU SEE

Treatment decision 'IV vitamin K — first line'

WHAT IT ENCODES

For vitamin K deficiency or an elevated INR, vitamin K (phytonadione) is first-line; the route depends on urgency: IV for serious situations (give slowly — anaphylactoid risk), oral for non-urgent correction (well absorbed if no cholestasis). It restores carboxylation but takes hours. Mnemonic: 'replace the cofactor first.'

BOARD TRAP

Vitamin K works SLOWLY (~6–12 h to make new functional factors) — it is NOT adequate for emergent reversal on its own. For a life-threatening bleed, give factor replacement (4F-PCC) PLUS vitamin K. Subcutaneous dosing is erratic and not preferred for reversal.

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10. Life-threatening bleed → 4F-PCC

WHAT YOU SEE

Reversal armory '4F-PCC, small volume, fast, life-threatening bleed or emergency surgery'

WHAT IT ENCODES

Major/life-threatening bleeding (e.g., intracranial hemorrhage) or emergency surgery on warfarin: give 4-factor prothrombin complex concentrate (4F-PCC — contains II, VII, IX, X) for RAPID, low-volume reversal, PLUS IV vitamin K. 4F-PCC normalizes the INR within minutes. Mnemonic: 'PCC for the panic — fast and small-volume.'

BOARD TRAP

PCC reverses fast but its effect is TRANSIENT (factors decay in hours) — you MUST co-administer IV vitamin K to sustain correction, or the INR rebounds. PCC carries a thrombotic risk.

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11. FFP when PCC unavailable

WHAT YOU SEE

Top-right 'FFP + VK when 4F-PCC unavailable, TACO risk'

WHAT IT ENCODES

Fresh frozen plasma (FFP) + vitamin K is the FALLBACK for warfarin reversal when 4F-PCC is unavailable. FFP supplies all factors but requires LARGE volumes (~15 mL/kg), is slower (thaw + infuse), and risks volume overload/TACO and TRALI. Mnemonic: 'FFP is the backup — big, slow, wet.'

BOARD TRAP

FFP is NOT first choice when PCC is available — it is slower, volume-heavy (TACO/TRALI risk), and gives less reliable, less rapid correction than 4F-PCC.

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12. Non-emergent INR → oral VK

WHAT YOU SEE

'Non-emergent bleed or elective surgery, restrict factors within 8-10h, correction continuing'

WHAT IT ENCODES

Non-emergent setting — minor bleed or elective surgery on warfarin: HOLD warfarin and give oral (low-dose) vitamin K; the INR corrects over ~8–24 h. No PCC/FFP needed because there's time. Hold warfarin ~5 days before elective surgery; bridge only if high thrombotic risk. Mnemonic: 'no rush → hold + oral K.'

BOARD TRAP

Do NOT over-reverse an elevated INR in the absence of serious bleeding — aggressive reversal/large vitamin K doses risk REBOUND thrombosis and make re-anticoagulation difficult.

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13. Asymptomatic high INR

WHAT YOU SEE

Bottom-right 'asymptomatic, low/no-dose VK, 1mg oral or IV, lowers bleeding risk'

WHAT IT ENCODES

Asymptomatic elevated INR on warfarin: for INR up to ~10 with no bleeding, simply HOLD warfarin ± low-dose oral vitamin K (~1–2.5 mg); for INR >10 with no bleeding, hold and give low-dose oral vitamin K (~2.5–5 mg). Routine vitamin K for mildly elevated INR (<4.5–5) is generally unnecessary. Mnemonic: 'no bleed → hold ± a little K.'

BOARD TRAP

LARGE vitamin K doses for an asymptomatic high INR cause prolonged WARFARIN RESISTANCE, making it hard to re-establish therapeutic anticoagulation. Use low doses, and do not transfuse plasma for a non-bleeding patient.
